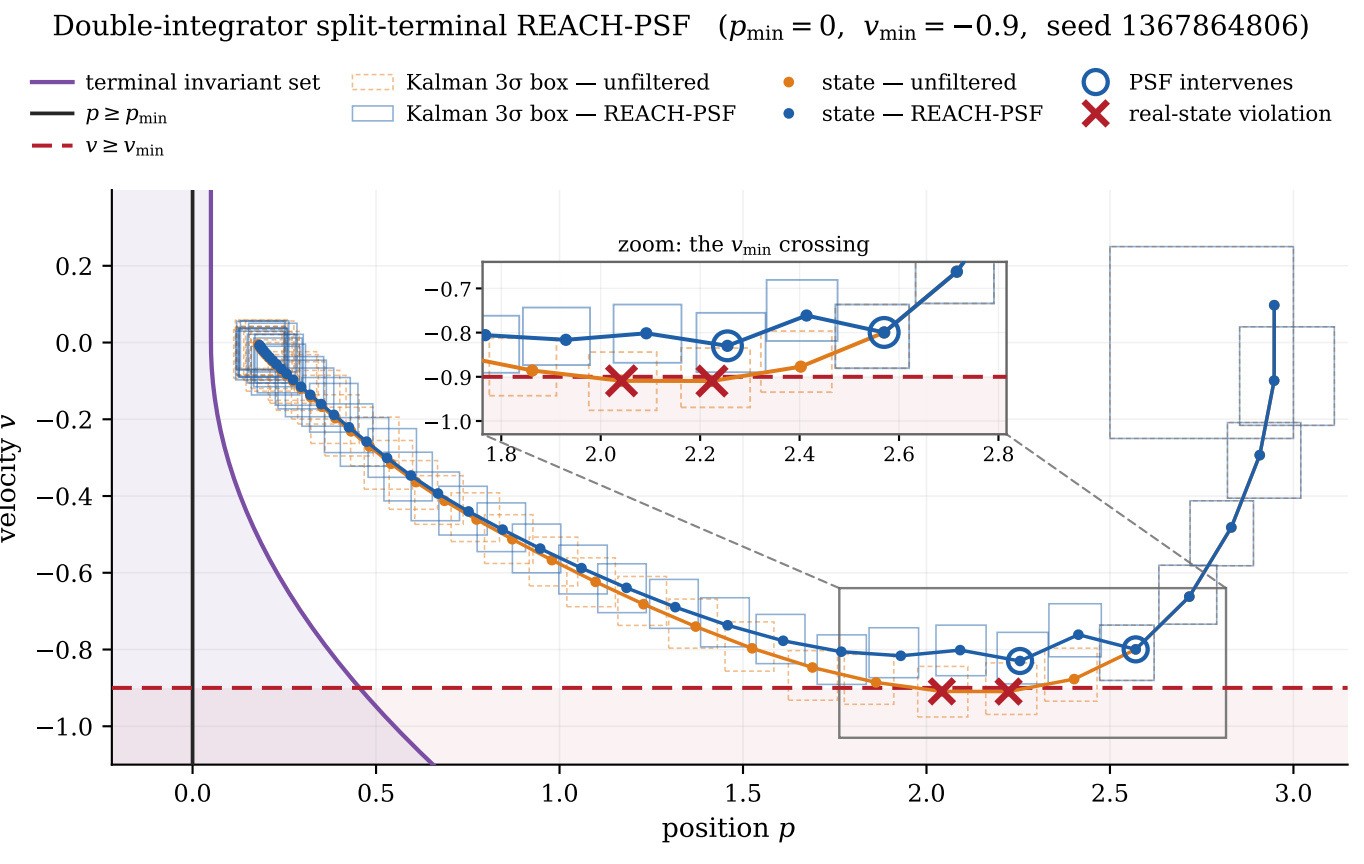

Double-integrator split-terminal REACH-PSF ($p_{\min} = 0$, $v_{\min} = -0.9$, seed 1367864806)
terminal invariant set
$p \geq p_{\min}$
$v \geq v_{\min}$
Kalman 3σ box — unfiltered
Kalman 3σ box — REACH-PSF
state — unfiltered
state — REACH-PSF
PSF intervenes
real-state violation
zoom: the $v_{\min}$ crossing
velocity v
position p
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
−0.7
−0.8
−0.9
−1.0
1.8
2.0
2.2
2.4
2.6
2.8